

Local entangling gates preserve an energy gap uniformly in chain length

The standard cluster chain gives a local interacting quantum Hamiltonian with an entangled unique ground state and excitation gap $J > 0$ for every open chain of $N \geq 2$ spins, when boundary terms are retained. The mechanism is exact unitary conjugation of independent spins. It supplies an energy gap directly; quantum kinematics and the action factor converting energy into frequency are inputs. This is an established construction worked through for G05.

1. Specify the Hamiltonian and the local circuit

The Hilbert space is $\mathcal{H}_N = (\mathbb{C}^2)^{\otimes N}$, with its full finite-dimensional operator domain. Dimensionless Pauli matrices X_j, Z_j act at site j . Fix a lattice spacing $a > 0$ and an energy $J > 0$, independent of N . Start with

$$H_N^{(0)} = \frac{J}{2} \sum_{j=1}^N (I - X_j), \quad U_N = \prod_{j=1}^{N-1} \text{CZ}_{j,j+1}, \quad H_N = U_N H_N^{(0)} U_N^\dagger.$$

On computational bits, $\text{CZ}|b, c\rangle = (-1)^{bc}|b, c\rangle$. All gates commute, are Hermitian and square to identity. Odd bonds form one layer of disjoint gates and even bonds a second, so circuit depth is at most two for every open chain length. This is a specification of a model Hamiltonian; the circuit alone is not a laboratory implementation of it.

Conjugation gives $\text{CZ}_{j,k} X_j \text{CZ}_{j,k} = X_j Z_k$: flipping bit b_j changes the gate phase by $(-1)^{b_k}$. Hence

$$K_1 = X_1 Z_2, \quad K_N = Z_{N-1} X_N, \quad K_j = Z_{j-1} X_j Z_{j+1} \quad (2 \leq j \leq N-1),$$

and

$$H_N = \frac{J}{2} \sum_j (I - K_j).$$

Each term has norm J and support of diameter at most $2a$. The bulk contains three-site interactions in the specified physical spin tensor factors. Commutation and independence of the K_j follow by conjugating the independent X_j ; in particular, no hidden product relation removes single-defect states. Nonlocal entangling conjugation preserves spectrum without making the physical-site ground state a product.

2. Ground state, entanglement and all excitation energies

For $s_j \in \{+1, -1\}$ let $|s\rangle_X$ be the product X eigenbasis. The vectors $U_N |s\rangle_X$ are a complete orthonormal eigenbasis and

$$E(s) = \frac{J}{2} \sum_j (1 - s_j) = J \# \{j : s_j = -1\}.$$

Thus the unique ground state is $|C_N\rangle = U_N |+\rangle^{\otimes N}$, $E_0 = 0$, the eigenvalue mJ has multiplicity $\binom{N}{m}$, and

$$\Delta_N = \min(\text{spec } H_N \setminus \{0\}) = J, \quad \inf_{N \geq 2} \Delta_N = J.$$

The state $Z_j |C_N\rangle$ is a normalized energy- J eigenstate: Z_j flips precisely the K_j sign. This checks attainment, not just a lower bound.

Entanglement can be seen at any cut between sites k and $k+1$. Factor U_N into gates internal to each side and the single crossing CZ gate. Internal gates are local unitaries relative to the cut. On the crossing pair,

$$\text{CZ}|+, +\rangle = \frac{|0\rangle|+\rangle + |1\rangle|-\rangle}{\sqrt{2}}.$$

The two Schmidt coefficients are $1/\sqrt{2}$, with all other spins initially factorized. Every such cut therefore has entanglement entropy $\log 2$ (natural logarithm). Interaction and entanglement coexist with a spectrum that is exactly as simple as the independent-spin spectrum.

3. Boundary and scale tests

Removing the endpoint terms gives, for $N \geq 3$,

$$H_N^{\text{bulk}} = \frac{J}{2} \sum_{j=2}^{N-1} (I - K_j) = U_N \left[\frac{J}{2} \sum_{j=2}^{N-1} (I - X_j) \right] U_N^\dagger.$$

The endpoint X labels are now unconstrained. The ground space is four-dimensional, but the gap *above that entire space* remains J . The separation of the two lowest eigenvalues counted with multiplicity is zero. Boundary conventions therefore matter for a unique-vacuum claim. For $N = 2$ this bulk-only operator is zero and has no positive excitation.

For an even periodic chain $N \geq 4$, include the closing CZ gate and all N cyclic stabilizers. The two-layer construction, independent eigenlabels, unique ground state and gap J persist. No symmetry constraint is imposed on the individual circuit gates here; no conclusion about symmetry-preserving phase equivalence is needed.

The volume limit in this note is $N \rightarrow \infty$ at fixed a, J : a uniform finite-volume excitation-gap statement. No continuum limit $a \rightarrow 0$, infinite-volume operator construction or relativistic dispersion is asserted. The defects in this exactly commuting model have no hopping term. If instead $J = J_N \rightarrow 0$, the gap closes; replacing H_N by H_N/N also gives J/N . Locality and entanglement do not choose a positive energy normalization.

4. The action parameter and the strategic consequence

To specify physical unitary time evolution, supply an action constant $\kappa > 0$:

$$i\kappa \partial_t |\psi\rangle = H_N |\psi\rangle, \quad \omega_{\text{excitation}} = J/\kappa.$$

This positive excitation frequency is relative to the ground state; it is not a relaxation rate. Unitary evolution does not equilibrate arbitrary states. The spectrum was defined in energy units without a Markov sampling clock, while its conversion to time still needs κ (normally $\hbar$). The construction does not select quantum kinematics or a universal action unit.

G05 closes the requested direct-Hamiltonian construction. A consequential next test is whether an extensive local perturbation that does not commute with the stabilizers preserves a volume-uniform gap. A concrete candidate is $-g \sum_j Z_j Z_{j+1}$: CZ conjugation leaves this perturbation unchanged, so its competition with the cluster terms reduces exactly to an Ising chain in a transverse field. That test addresses robustness beyond the exact conjugation family; it is a separate task, not a claim proved here.

Source and verification scope

Seifnashri and Shao, *Cluster state as a non-invertible symmetry protected topological phase*, arXiv:2404.01369v2, Introduction Eqs. (1)–(3), supplies the periodic cluster Hamiltonian, its

unique gapped ground state and the controlled-Z entangler. The source normalizes the Hamiltonian as $-\sum_j K_j$; ours adds $NJ/2$ times identity and multiplies by $J/2$, so its dimensionless single-defect cost 2 becomes energy J . Reading coverage: that introductory passage only, checked as primary HTML on 2026-09-14; later symmetry results were not audited. One discovery query and one primary-page read suffice for this established-model reuse.

The written checks above cover conjugation, complete eigenbasis, gap attainment, Schmidt coefficients and missing-boundary degeneracy. No new ledger claim, novelty claim or computational verification is introduced.